$$= \left(e\log e - 1\log 1\right) - \left[x\right]_1^e$$
$$= \left(e-0\right) - \left(e-1\right)$$
$$= 1$$

Example 2.70

Evaluate $\int_0^{\frac{\pi}{2}} x \sin x \, dx$

Solution:

Take $u = x$	and $dv = \sin x \, dx$
Differentiate	Integrate
$du = dx$	$v = -\cos x$

$$\int_0^{\frac{\pi}{2}} x \sin x \, dx = \int_0^{\frac{\pi}{2}} u dv$$
$$= \left[uv\right]_0^{\frac{\pi}{2}} - \int_0^{\frac{\pi}{2}} v du$$
$$= \left[-x\cos x\right]_0^{\frac{\pi}{2}} + \int_0^{\frac{\pi}{2}} \cos x \, dx$$
$$= 0 + \left[\sin x\right]_0^{\frac{\pi}{2}} = 1$$

Example 2.71

If $\int_1^a 3x^2 \, dx = -1$, then find the value of a, $a \in R$.

Solution:

Given that $\int_1^a 3x^2 \, dx = -1$

$$\left[x^3\right]_1^a = -1$$
$$a^3 - 1 = -1$$
$$a^3 = 0 \Rightarrow a = 0$$

Example 2.72

If $\int_a^b dx = 1$ and $\int_a^b x \, dx = 1$, then find a and b

Solution:

Given that $\int_a^b dx = 1$

$$\left[x\right]_a^b = 1$$
$$b - a = 1 \quad \ldots (1)$$

Now, $\int_a^b x \, dx = 1$

$$\left[\frac{x^2}{2}\right]_a^b = 1$$
$$b^2 - a^2 = 2$$
$$\left(b+a\right)\left(b-a\right) = 2$$
$$b + a = 2 \ldots (2) \quad \left[\because b - a = 1\right]$$
$$(1)+(2) \Rightarrow \quad 2b = 3$$
$$\therefore b = \frac{3}{2}$$

Now, $\frac{3}{2} - a = 1 \quad \left[\because \text{from}(1)\right]$

$$\therefore a = \frac{1}{2}$$

Example 2.73

Evaluate $\int_1^4 f\left(x\right) dx$, where

$$f\left(x\right) = \begin{cases} 7x+3, & if \ 1 \le x \le 3 \\ 8x, & if \ 3 \le x \le 4 \end{cases}$$

Solution:

$$\int_1^4 f\left(x\right) dx = \int_1^3 f\left(x\right) dx + \int_3^4 f\left(x\right) dx$$
$$= \int_1^3 \left(7x+3\right) dx + \int_3^4 8x \, dx$$
$$= \left[\frac{7x^2}{2} + 3x\right]_1^3 + \left[\frac{8x^2}{2}\right]_3^4$$
$$= \frac{63}{2} + 9 - \frac{13}{2} + 64 - 36$$
$$= 62$$

Example 2.74

If $f(x) = \begin{cases} x^2, & -2 \le x < 1 \\ x, & 1 \le x < 2 \\ x-4, & 2 \le x \le 4 \end{cases}$,

then find the following

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